

Design and Implementation of an EMG-Based Robotic Arm Using ESP32

Project Report

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Declaration

We hereby declare that the project report we submitted was written as a part of our degree program at Jatiya Kabi Kazi Nazrul Islam University .We further declare that this project work has not been submitted to any other university or institution for the award of any degree, diploma, or other qualification. The information presented in this report is true and correct to the best of my knowledge and belief.

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Approval

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Abstract

This project presents the design and implementation of an EMG-based robotic arm using an ESP32 microcontroller, where human muscle signals are used to control robotic hand movement. Surface EMG signals are acquired, processed, and converted into PWM signals to drive servo motors, enabling natural and real-time gripping actions. The system offers a low-cost and intuitive solution for assistive, rehabilitation, and educational applications.

Table of Contents

Section	Title	Page No.
Chapter 1	INTRODUCTION	1
1.1	Background	1
1.2	Project Motivation	2
1.3	Objectives of the Project	2
1.4	Project Overview	3
Chapter 2	LITERATURE REVIEW	5
Chapter 3	SYSTEM DESIGN AND METHODOLOGY	7
3.1	Hardware Components Used	7
3.2	Block Diagram Description	11
3.3	Software Design	13
3.4	Working Principle	16
Chapter 4	REAL LIFE ROLE AND FUTURE SCOPE	18
4.1	Real Life Applications	18
4.2	Advantages	19
4.3	Limitations	20
4.4	Future Scope	20
Chapter 5	RESULT AND CONCLUSION	22
5.1	Result	22
5.2	Discussion	23
5.3	Conclusion	24

List of Figures

Figure No.	Figure Title	Page No.
Fig. 1	EMG Muscle Sensor With Cable And Electrodes Controller	8
Fig. 2	ESP-32	9
Fig. 3	Servo Motor	10
Fig. 4	3.7 and 9 V battery	10
Fig. 5	Block diagram of circuit of the system	12
Fig. 6	Hardware Implementation of EMG Based Robotic Arm System	13
Fig. 7	Flowchart of ESP-32 code	15
Fig. 8	Block diagram of working principle of EMG Controlled Robotic Arm	17

List of Tables

Table No.	Table Title	Page No.
Table 1	Relationship Between EMG Signal Strength and Servo Motor Angle	22

Chapter 1: Introduction

1.1 Background

In recent years, rapid development in the fields of Electrical and Electronic Engineering, biomedical engineering, and robotics has led to the emergence of intelligent systems that are capable of interacting directly with human biological signals. One of the most significant advancements in this area is the use of bio-signals to control electronic and mechanical systems. Among various bio-signals such as Electroencephalography (EEG), Electrocardiography (ECG), and Electrooculography (EOG), Electromyography (EMG) has gained special attention due to its direct relationship with muscle activity.

Electromyography (EMG) is the technique used to measure and analyze the electrical signals produced by skeletal muscles during contraction and relaxation. These electrical signals are generated due to the flow of ions across muscle fiber membranes when muscles are activated by the nervous system. EMG signals provide valuable information regarding muscle strength, movement intention, and neuromuscular health. Because of this property, EMG signals are widely used in medical diagnosis, rehabilitation engineering, prosthetics, sports science, and human-machine interface applications.

Robotic arms are mechanical devices designed to mimic the movements and functions of the human arm. Traditionally, robotic arms are controlled using predefined programs, switches, joysticks, or remote controllers. While these methods are effective, they lack natural interaction and intuitive control. To overcome this limitation, researchers and engineers have started integrating biological signals with robotic systems, enabling machines to respond directly to human intention.

An EMG-controlled robotic arm represents a significant step toward natural and intelligent control systems. By using EMG signals, the robotic arm can respond directly to muscle contractions, making the control system more user-friendly and realistic. Such systems are especially useful in assistive technologies, where physically challenged or amputee individuals can control prosthetic limbs using their own muscle signals.

With the advancement of microcontroller technology, compact and powerful controllers like the ESP-32 have made it possible to implement complex signal processing and motor control in low-cost embedded systems. The ESP-32 offers high processing speed, built-in analog-to-digital converters (ADC), and PWM outputs, making it suitable for EMG-based control applications.

This project, titled "EMG Controlled Robotic Arm by ESP-32", focuses on designing and implementing a functional robotic hand that can grip objects based on muscle activity detected through an EMG sensor. The project integrates concepts from biomedical signal acquisition,

embedded systems, motor control, and mechanical design, making it a multidisciplinary engineering application.

1.2 Project Motivation

The primary motivation behind this project arises from the increasing demand for intelligent assistive devices and human-machine interaction systems. In modern society, there is a growing need for technologies that can improve the quality of life for people with physical disabilities, muscle weakness, or limb loss. Prosthetic limbs controlled by natural muscle signals offer a promising solution to this challenge.

Conventional prosthetic and robotic systems often rely on complex control mechanisms, which may be expensive, difficult to operate, and require extensive training. In contrast, EMG-based control systems allow users to control devices using their natural muscle movements, making the system intuitive and easy to use. This inspired the idea of developing a simple yet effective EMG-controlled robotic arm using easily available components.

From an academic perspective, this project provides an excellent opportunity to apply theoretical knowledge gained in courses such as Analog Electronics, Digital Electronics, Biomedical Instrumentation, Microcontrollers, and Control Systems. Instead of limiting learning to simulations and theoretical analysis, this project emphasizes hands-on experience with real-world signals and hardware.

Another important motivation is cost efficiency. Many commercial EMG-controlled prosthetic systems are highly expensive and inaccessible to people in developing countries. By using low-cost components such as an ESP-32 microcontroller, standard servo motors, and a 3D printed robotic hand, this project aims to demonstrate that functional EMG-based robotic systems can be developed at a much lower cost.

Additionally, this project is motivated by the desire to explore emerging technologies such as wearable sensors, embedded intelligence, and smart robotics. It encourages innovation and creativity while addressing real-life engineering problems. The project also serves as a foundation for future research and development in advanced prosthetics and rehabilitation devices.

1.3 Objectives of the Project

The primary objective of this project is to design and implement an EMG controlled robotic arm capable of performing gripping actions by interpreting human muscle activity. The project focuses on developing a simple, reliable, and low-cost system that can translate electromyographic signals into mechanical movement.

The specific objectives of the project are as follows:

1. To acquire and process human muscle signals using an EMG sensor and convert them into usable control signals.
2. To interface the EMG sensor with the ESP-32 microcontroller for real-time signal processing and generation of PWM signals for servo motor control.
3. To design and develop a 3D printed robotic hand that can perform gripping and releasing actions based on muscle contraction.

1.4 Project Overview

The "EMG Controlled Robotic Arm by ESP-32" project is designed to convert human muscle signals into mechanical movement through an embedded control system. The system consists of sensing, processing, control, and actuation stages, all working together to achieve smooth and responsive operation.

The sensing stage involves the EMG sensor, which is attached to the user's muscle using surface electrodes. When the user contracts the muscle, electrical signals are generated due to neuromuscular activity. These weak signals are captured by the EMG sensor, amplified, filtered, and converted into an analog voltage signal.

In the processing stage, the analog output from the EMG sensor is fed into the analog input pin of the ESP-32 microcontroller. The ESP-32 continuously reads the EMG signal using its built-in ADC. The digital values obtained from the ADC represent the intensity of muscle activity. Software-based filtering and threshold comparison are applied to distinguish between relaxed and contracted muscle states.

The control stage involves mapping the processed EMG signal values to appropriate servo motor angles. When the muscle contraction exceeds a predefined threshold, the ESP-32 generates PWM signals to rotate the servo motors. The rotation angle of the servo motor depends on the strength of the muscle contraction.

The actuation stage consists of two servo motors mechanically connected to a 3D printed robotic hand. These servo motors pull or release strings or linkages connected to the fingers, enabling the hand to open and close. As a result, the robotic hand performs gripping actions similar to a human hand.

Power management is an important aspect of the system. Four 3.7V rechargeable batteries connected in parallel are used to supply sufficient current to the ESP-32 and servo motors. Two separate 9V batteries are used to power the EMG sensor to minimize noise and interference caused by motor operation.

Overall, this project demonstrates a complete embedded system that integrates biomedical signal acquisition with robotic control. It serves as a practical example of how human physiological signals can be effectively used to control machines, paving the way for advanced assistive and prosthetic technologies.

Chapter 2: Literature Review

Literature Review

In recent years, the field of assistive robotics and biomedical engineering has gained significant attention due to the increasing demand for intelligent systems that can support individuals with physical disabilities and motor impairments. One of the most promising approaches in this domain is the use of electromyography (EMG) signals for controlling robotic and prosthetic devices. EMG signals are generated as a result of electrical activity in muscles during contraction and relaxation, and these signals provide valuable information about human intention and movement.

Several research works have explored EMG-based control systems for robotic arms and prosthetic hands. Early robotic arm control systems primarily relied on mechanical switches, joysticks, or pre-programmed motion sequences. Although these methods were simple to implement, they lacked natural control and required continuous manual effort from the user. As a result, researchers shifted their focus toward bio-signal-based control methods, where human physiological signals are directly used to operate machines.

EMG-based robotic control systems have been widely studied due to their ability to provide intuitive and real-time interaction between humans and machines. In many previous studies, surface EMG sensors were used to capture muscle signals from the forearm or upper arm. These signals were then amplified, filtered, and processed using microcontrollers or digital signal processing units. Based on the processed EMG data, robotic arms were able to perform actions such as gripping, lifting, and releasing objects. Such systems demonstrated improved usability compared to traditional control methods.

A number of studies have implemented EMG controlled robotic arms using microcontrollers such as Arduino Uno, Arduino Mega, and PIC microcontrollers. These systems generally used threshold-based techniques to differentiate between muscle contraction and relaxation. While these implementations were effective, many of them suffered from limitations such as low processing speed, limited ADC resolution, and lack of advanced features. Additionally, some systems required complex external circuits to process EMG signals, increasing system complexity and cost.

Recent literature shows a growing interest in using more advanced microcontrollers and embedded platforms to improve system performance. High-performance microcontrollers with better ADC resolution, higher clock speed, and integrated communication modules have enabled more efficient EMG signal processing. Researchers have also explored wireless EMG-based control systems to enhance portability and user comfort. However, many of these systems remain expensive or too complex for educational and low-cost applications.

Another important area discussed in previous research is the mechanical design of robotic hands. Several robotic hands have been developed using metal structures or complex mechanical assemblies, which increase weight and cost. In contrast, 3D printing technology has emerged as a

cost-effective solution for developing lightweight and customizable robotic hands. Many studies have successfully demonstrated the use of 3D printed robotic hands controlled by servo motors, offering sufficient grip strength for lightweight objects while maintaining simplicity in design.

Despite the progress in EMG-based robotic systems, certain challenges are repeatedly highlighted in the literature. EMG signals are highly sensitive to noise, electrode placement, muscle fatigue, and user variability. Improper electrode placement can significantly affect signal quality and system performance. Additionally, systems with limited degrees of freedom often fail to provide precise finger-level control, which restricts their application in advanced prosthetic devices. These limitations indicate the need for improved signal processing techniques, better hardware design, and user-friendly system architecture.

Based on the review of existing works, it is evident that there is a strong need for a low-cost, simple, and reliable EMG controlled robotic arm that can be used for educational, rehabilitation, and assistive purposes. Most existing systems either focus on high-end prosthetic solutions or remain limited to basic laboratory demonstrations. Therefore, the motivation behind this project is to bridge the gap by developing an EMG controlled robotic arm using a powerful yet affordable microcontroller, simple signal processing techniques, and a lightweight 3D printed robotic hand.

The present work builds upon the fundamental concepts discussed in previous studies while introducing improvements in system simplicity, cost effectiveness, and practical implementation. By using an ESP-32 microcontroller, separating power supplies to reduce noise, and employing a threshold-based control strategy, this project aims to provide a stable and intuitive EMG-based robotic control system suitable for real-world applications and academic learning.

Chapter 3: System Design and Methodology

3.1 Hardware Components Used

This section describes each hardware component used in the project, along with its working principle and role in the system.

3.1.1 EMG Sensor

The EMG sensor is one of the most critical components of the system, as it directly interacts with the human body. The EMG sensor is designed to detect electrical signals produced by skeletal muscles during contraction.

When a muscle contracts, motor neurons send electrical impulses to muscle fibers, causing ion exchange across cell membranes. This generates electrical signals known as EMG signals, which typically have very low amplitude, usually in the range of microvolts (μV).

To make these signals usable, the EMG sensor includes:

- An instrumentation amplifier to amplify the weak signals
- High-pass and low-pass filters to remove noise
- A reference electrode for signal stabilization

Surface electrodes are placed on the skin above the target muscle. Proper electrode placement is essential to obtain accurate and noise-free EMG signals. Usually, two electrodes are placed on the muscle, and one electrode is used as a reference or ground.

The output of the EMG sensor is an analog voltage signal proportional to the intensity of muscle contraction. This voltage signal is suitable for interfacing with the analog input pin of the ESP-32.

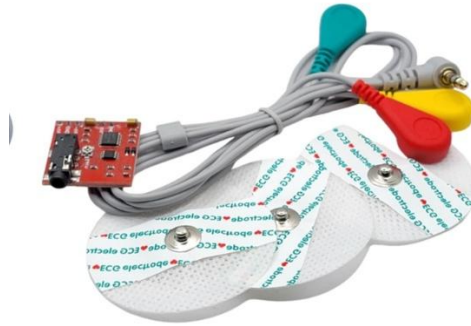


Figure : EMG Muscle Sensor With Cable And Electrodes Controller
[<https://store.roboticsbd.com>]

3.1.2 ESP-32 Microcontroller

The ESP-32 microcontroller serves as the central processing unit of the entire system. It is a high-performance microcontroller with dual-core processing capability, multiple ADC channels, PWM outputs, and built-in Wi-Fi and Bluetooth modules.

In this project, the ESP-32 performs several important tasks:

- Reading analog EMG signals using ADC
- Processing EMG data in real time
- Generating PWM signals for servo control
- Managing overall system operation

The built-in ADC of the ESP-32 converts the analog EMG voltage into digital values. These digital values represent muscle activity levels and are processed by the microcontroller software.

The ESP-32 is programmed using the Arduino IDE, which provides a simple and efficient platform for embedded system development. The microcontroller was chosen due to its high speed, low cost, and suitability for real-time biomedical applications.



Figure: ESP-32[<https://store.roboticsbd.com>]

3.1.3 Servo Motors

Servo motors are electromechanical actuators used to produce precise angular motion. In this project, two servo motors are used to control the movement of five fingers of the robotic hand.

Servo motors are selected because of the following advantages:

- Accurate position control
- Simple PWM-based interfacing
- High torque at low speed
- Compact size and low power consumption

Each servo motor rotates to a specific angle based on the PWM signal generated by the ESP-32. The rotation angle corresponds to the strength of the muscle contraction detected by the EMG sensor.

The servo motors are mechanically connected to the fingers of the robotic hand using strings or linkage mechanisms. When the servo motor rotates, it pulls or releases the strings, causing the fingers to bend or straighten.



Figure : Servo Motor [<https://store.roboticsbd.com>]

3.1.4 Power Supply Unit

A stable and reliable power supply is essential for proper system operation. In this project, separate power sources are used for different modules to improve performance and reduce noise.

- Four 3.7V rechargeable batteries connected in parallel are used to power the ESP-32 and servo motors. The parallel configuration increases current capacity, which is required for servo motor operation.
- Two 9V batteries are used exclusively to power the EMG sensor. This separation minimizes electrical noise caused by servo motor switching and improves signal accuracy.

Proper grounding between modules is maintained to ensure safe and stable operation.



Figure : 3.7V and 9V battery [<https://store.roboticsbd.com>]

3.1.5 3D Printed Robotic Hand

The 3D printed robotic hand forms the mechanical output stage of the system. It is designed to replicate the basic gripping function of a human hand.

3D printing technology is used because it offers:

- Low fabrication cost
- Lightweight structure
- Design flexibility
- Easy customization

The fingers of the robotic hand are connected to the servo motors through strings or mechanical linkages. When the servo motor rotates, the fingers move accordingly, allowing the hand to grip or release objects.

The design focuses on simplicity and functionality, making it suitable for educational, experimental, and assistive applications.

3.2 Block Diagram Description

The block diagram represents the overall structure and functional relationship between different components of the EMG controlled robotic arm system. It provides a clear overview of how the system operates from input signal acquisition to final mechanical output.

The major blocks of the system are:

- EMG Sensor
- ESP-32 Microcontroller
- Servo Motors
- Power Supply Unit
- 3D Printed Robotic Hand

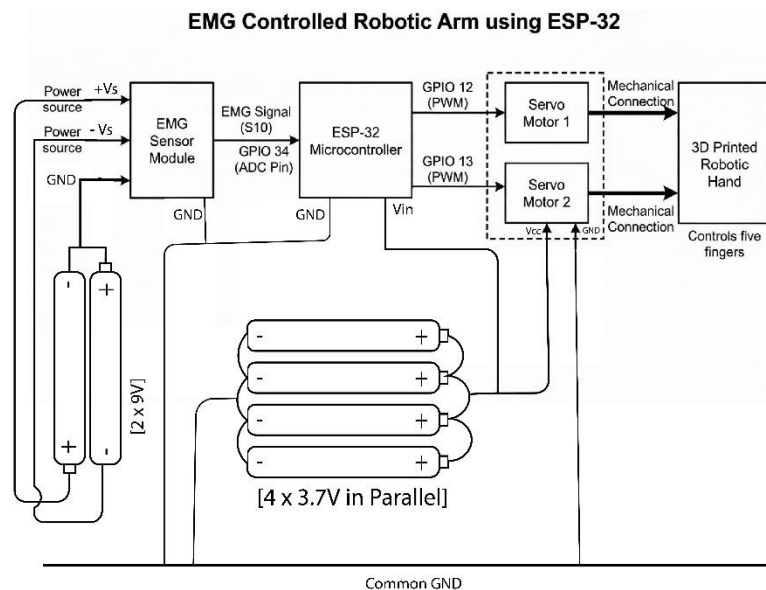


Fig : Block diagram of circuit of the system

The operation of the system begins with the EMG sensor, which is responsible for capturing the electrical signals generated by muscle contraction. When a user contracts a muscle, small electrical potentials are produced due to neuromuscular activity. These signals are extremely weak and cannot be directly processed by a microcontroller.

The EMG sensor amplifies and filters these weak bio-signals and converts them into a usable analog voltage. This analog voltage is then sent to the ESP-32 microcontroller, which acts as the brain of the system.

The ESP-32 continuously reads the EMG signal through its analog-to-digital converter (ADC). The microcontroller processes the signal by applying thresholding and basic filtering techniques to distinguish between relaxed and contracted muscle states. Based on the processed signal strength, the ESP-32 generates Pulse Width Modulation (PWM) signals.

These PWM signals are sent to the servo motors, which rotate to specific angles according to the duty cycle of the PWM signal. The mechanical rotation of the servo motors is transferred to the 3D printed robotic hand, causing the fingers to open or close. As a result, the robotic hand performs gripping and releasing actions similar to a human hand.

The power supply unit provides the necessary electrical energy to all system components and ensures stable and reliable operation.

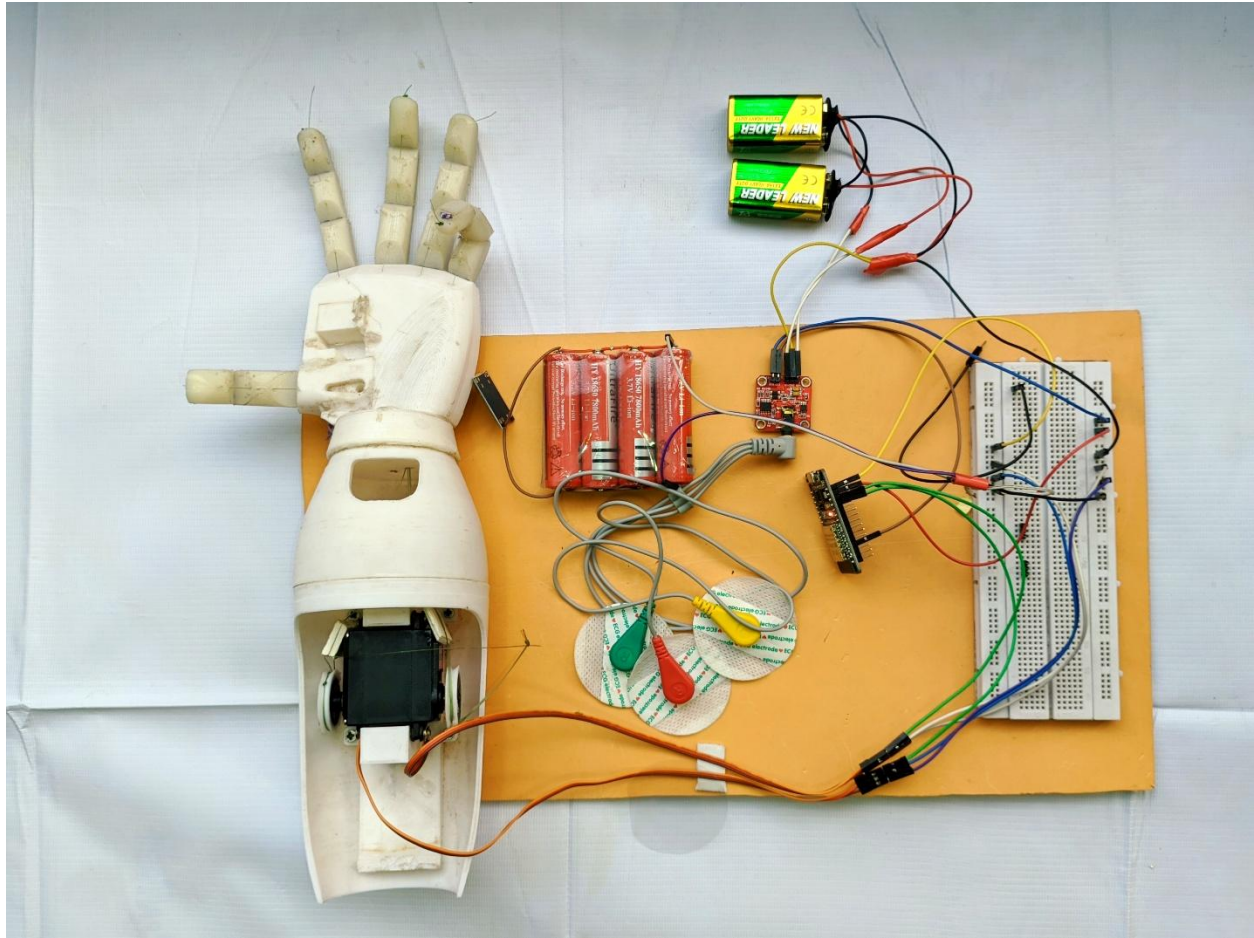


Fig : Hardware Implementation of EMG Based Robotic Arm System

3.3 Software Design

The software design plays a crucial role in the successful operation of the EMG controlled robotic arm. The software acts as the link between the human muscle signal acquired by the EMG sensor and the mechanical movement of the robotic hand. In this project, the software is responsible for reading the EMG signal, processing it, making logical decisions, and controlling the servo motors accordingly.

The software is developed using the Arduino Integrated Development Environment (Arduino IDE) and executed on the ESP-32 microcontroller. The program continuously monitors muscle activity in real time and converts it into corresponding servo motor movements.

3.3.1 Arduino IDE

The **Arduino Integrated Development Environment (Arduino IDE)** is used for writing, compiling, and uploading the program to the ESP-32 microcontroller. Arduino IDE is an open-source platform widely used for embedded system development due to its simplicity, flexibility, and extensive library support.

Arduino IDE provides:

- A user-friendly programming interface
- Easy integration with ESP-32 boards
- Built-in functions for analog reading and PWM control
- Support for servo motor and sensor libraries

In this project, the Arduino IDE is used to write the control algorithm in C/C++ language. The ESP-32 board is configured within the Arduino IDE by installing the ESP-32 board package. After configuration, the ESP-32 can be programmed similarly to other Arduino-compatible boards.

The main tasks performed by the software include:

- Initializing input and output pins
- Reading analog EMG sensor values using ADC
- Filtering raw EMG data to reduce noise
- Applying threshold logic to detect muscle contraction
- Generating PWM signals for servo motor control

The EMG sensor produces an analog voltage output that varies with muscle activity. This voltage is read using the **analogRead()** function of the ESP-32. Since EMG signals are noisy in nature, basic filtering such as moving average or smoothing techniques are applied in software to obtain stable readings.

A threshold value is defined in the program to distinguish between relaxed and contracted muscle states. When the EMG signal exceeds this threshold value, the ESP-32 interprets it as muscle contraction and sends appropriate PWM signals to the servo motors. As a result, the robotic hand performs gripping action. When the muscle relaxes, the servo motors return to their original position.

The Arduino IDE allows easy modification and tuning of parameters such as threshold values, delay time, and servo angle range, making the system flexible and customizable.

3.3.2 Code Flowchart and Algorithm Description

The code flowchart represents the logical sequence of operations performed by the ESP-32 microcontroller during system operation. It provides a clear understanding of how the software processes EMG signals and controls the robotic hand.

The software algorithm follows a continuous loop-based structure, ensuring real-time system response.

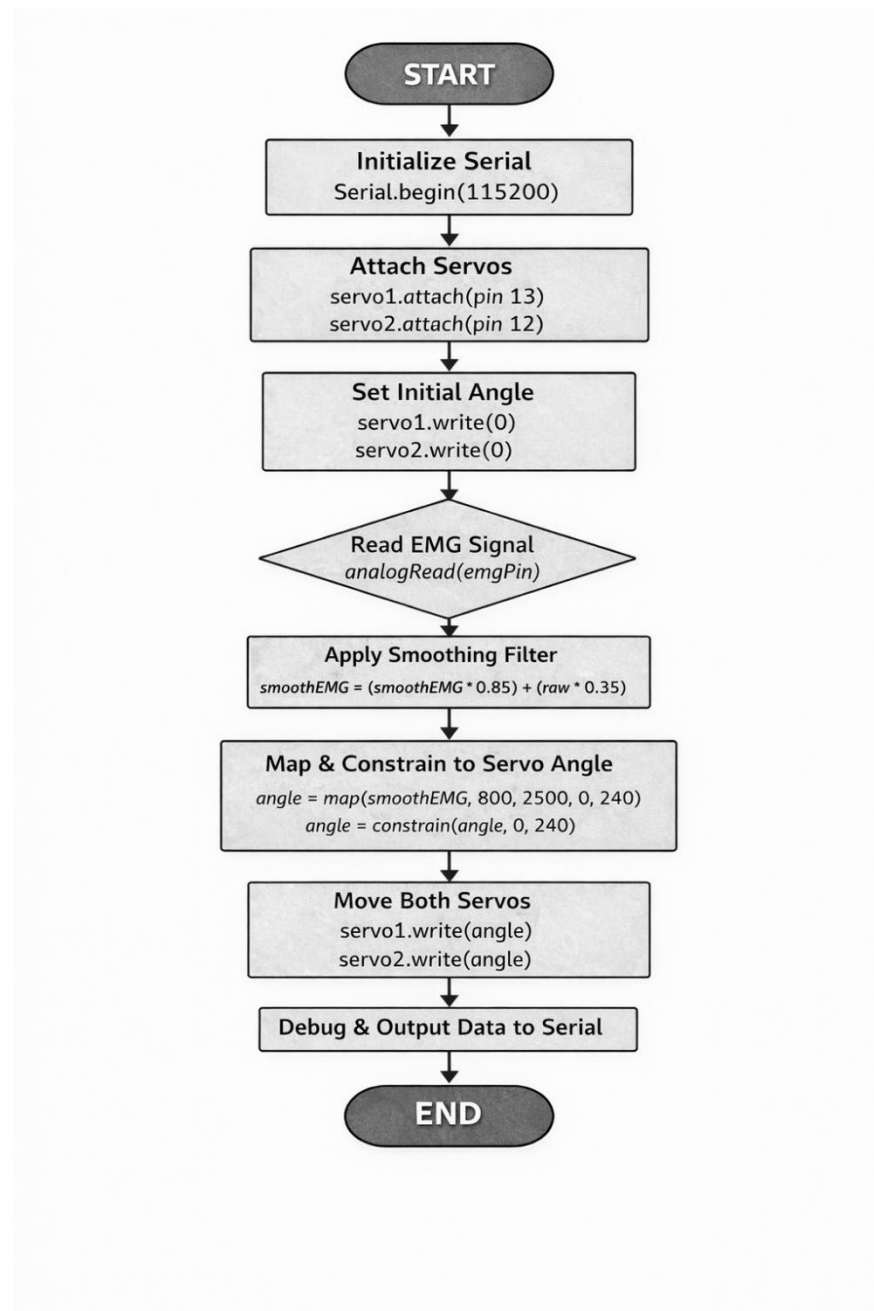


Figure : Flowchart of ESP-32 code

3.4 Working Principle

The working principle of the EMG controlled robotic arm is based on the acquisition of human muscle-generated electrical signals and their conversion into controlled mechanical movement through an embedded control system. This system establishes a direct interface between human muscular activity and robotic actuation, enabling natural and intuitive control of the robotic hand.

When a human muscle is activated, electrical impulses are generated due to the transmission of signals from motor neurons to muscle fibers. These impulses result in voltage variations on the surface of the skin above the muscle. The magnitude of these electrical signals increases with the strength of muscle contraction. These bio-electrical signals form the fundamental input to the system.

The EMG sensor, attached to the surface of the skin using electrodes, detects these weak muscle-generated electrical signals. Since the EMG signals are extremely low in amplitude and highly susceptible to noise, the EMG sensor incorporates amplification and filtering stages. The amplification stage increases the signal amplitude to a usable level, while the filtering stage removes unwanted noise components such as power-line interference and motion artifacts. The processed output of the EMG sensor is an analog voltage signal that varies in proportion to muscle activity.

This analog output voltage is fed to the analog input pin of the ESP-32 microcontroller. The ESP-32 continuously monitors the EMG signal using its built-in analog-to-digital converter (ADC). The ADC converts the incoming analog voltage into a digital value that represents the instantaneous level of muscle activity. These digital values are processed in real time by the microcontroller software.

The embedded program running on the ESP-32 analyzes the digitized EMG signal by applying threshold-based decision logic. A predefined threshold value is set to differentiate between relaxed and contracted muscle conditions. When the EMG signal level exceeds the threshold value, the system interprets it as intentional muscle contraction. When the signal remains below the threshold, the system assumes that the muscle is in a relaxed state.

Based on the processed EMG signal strength, the ESP-32 generates Pulse Width Modulation (PWM) control signals. These PWM signals are used to control the angular position of the servo motors. The duty cycle of the PWM signal determines the rotation angle of the servo motors, allowing proportional control of movement corresponding to muscle contraction intensity.

The servo motors convert the electrical PWM signals into precise mechanical rotational motion. This rotational motion is transmitted to the fingers of the 3D printed robotic hand through strings or mechanical linkage mechanisms. As a result, the fingers of the robotic hand open or close, enabling gripping and releasing actions. Stronger muscle contractions result in firmer gripping, while muscle relaxation causes the hand to return to its initial position.

The entire system operates continuously in a closed-loop manner, allowing real-time response to changes in muscle activity. The separation of power supplies for the EMG sensor and servo motors ensures minimal interference and stable signal processing. This working principle enables smooth, reliable, and intuitive control of the robotic hand using natural human muscle movements.

Working Principle: EMG Controlled Robotic Arm

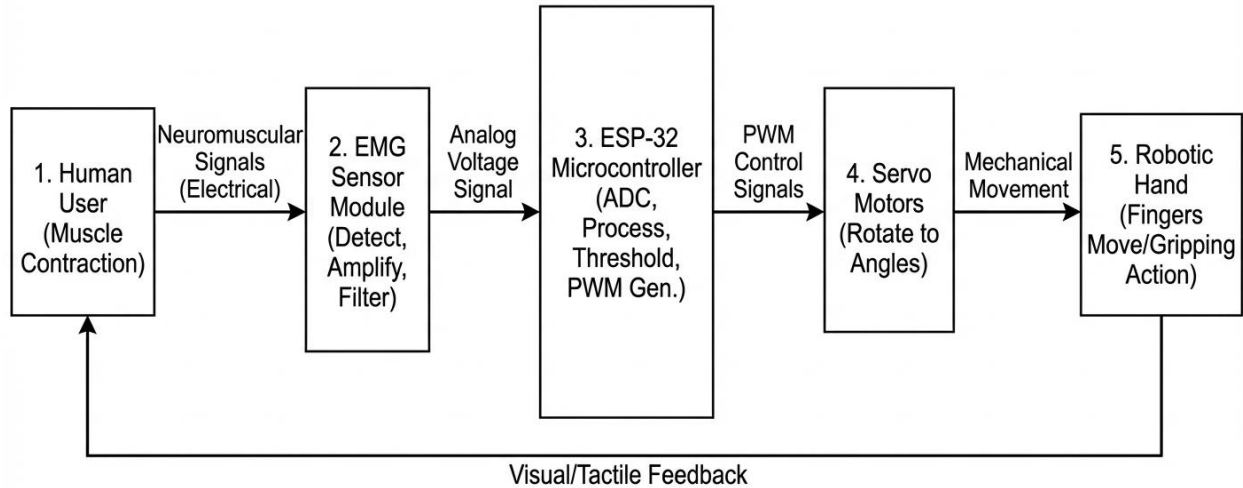


Fig: Block diagram of working principle of EMG Controlled Robotic Arm

Chapter 4: Real Life Role and Future Scope

4.1 Real Life Applications

The EMG controlled robotic arm has wide-ranging real-life applications due to its ability to translate human muscle activity into mechanical motion. Some of the major applications are discussed below.

Prosthetic Hand Control for Amputees

One of the most significant applications of EMG controlled robotic systems is in the field of prosthetics. For individuals who have lost a limb due to accident, disease, or congenital conditions, EMG-based prosthetic hands offer a natural and intuitive method of control. By placing EMG electrodes on the remaining muscles of the amputated limb, users can control a prosthetic hand using their own muscle contractions.

The EMG controlled robotic arm developed in this project demonstrates the basic principle used in modern prosthetic devices. With further enhancement, such systems can help amputees regain independence and improve their quality of life by enabling them to perform daily activities such as grasping objects, holding tools, and performing fine motor tasks.

Rehabilitation Systems for Patients with Muscle Weakness

EMG controlled robotic arms are widely used in rehabilitation engineering. Patients recovering from stroke, spinal cord injury, or neuromuscular disorders often suffer from muscle weakness or partial paralysis. Rehabilitation devices based on EMG signals can assist patients in performing controlled movements during therapy sessions.

The robotic arm can provide assistance proportional to the patient's muscle activity, encouraging active participation in rehabilitation. This type of therapy helps in muscle re-education, strength improvement, and faster recovery. The developed system can be adapted as a low-cost rehabilitation aid for physiotherapy applications.

Assistive Devices for Physically Challenged People

For individuals with physical disabilities, assistive robotic devices can significantly enhance daily living. EMG controlled robotic arms can be used as assistive tools to help users perform tasks that require gripping, lifting, or holding objects.

Since the control is based on muscle signals, the system offers a more natural interaction compared to conventional switch-based or joystick-controlled devices. This makes EMG-based assistive devices more user-friendly and accessible, especially for people with limited mobility.

Human–Robot Interaction Systems

Human–robot interaction (HRI) is an emerging research area focused on creating intuitive communication between humans and machines. EMG controlled robotic systems play an important role in this field by enabling robots to respond directly to human intentions.

The developed robotic arm demonstrates a basic form of HRI where human muscle activity directly controls robotic motion. Such systems can be extended for use in collaborative robots, industrial automation, and smart wearable technologies, where robots work alongside humans in a safe and efficient manner.

Educational and Research Platforms

The EMG controlled robotic arm also serves as an excellent educational and research platform. It provides hands-on experience in multidisciplinary fields such as biomedical engineering, embedded systems, robotics, and control engineering.

Students and researchers can use this system to study EMG signal characteristics, signal processing techniques, motor control algorithms, and human–machine interfaces. Due to its low cost and modular design, the system is well suited for academic laboratories and research projects.

4.2 Advantages

The EMG controlled robotic arm offers several advantages that make it suitable for practical implementation and further development.

Low-Cost and Simple Design

The system is designed using easily available and low-cost components such as ESP-32, standard servo motors, and a 3D printed robotic hand. This makes the project economically feasible and accessible, especially in developing countries.

Real-Time Response

The system operates in real time, allowing immediate response to muscle contraction. The ESP-32 processes EMG signals quickly and generates control signals without noticeable delay, ensuring smooth and responsive operation.

Intuitive Control Using Muscle Signals

Unlike traditional control methods, EMG-based control uses natural muscle movements. This makes the system intuitive and easy to use, as the user does not require extensive training to operate the robotic arm.

Compact and Portable System

The compact design and battery-powered operation make the system portable. This portability allows the robotic arm to be used in different environments such as hospitals, homes, and laboratories.

4.3 Limitations

Despite its advantages, the EMG controlled robotic arm also has certain limitations that need to be considered.

Signal Noise and Interference

EMG signals are highly sensitive to noise caused by improper electrode placement, muscle movement, and electrical interference. Noise can affect signal accuracy and lead to unreliable system performance.

Limited Degrees of Freedom

The current system uses two servo motors to control five fingers, which limits the degrees of freedom of the robotic hand. This restricts the range of possible movements and fine motor control.

Effect of Muscle Fatigue

Muscle fatigue affects the amplitude and consistency of EMG signals. Prolonged usage may result in reduced system accuracy and inconsistent control performance.

4.4 Future Scope

The future scope of the EMG controlled robotic arm is vast, and several improvements can be made to enhance system performance and functionality.

Use of Multiple EMG Sensors

By using multiple EMG sensors placed on different muscles, individual finger control can be achieved. This will significantly improve the dexterity and realism of the robotic hand.

Wireless Monitoring and Control

The built-in Wi-Fi and Bluetooth capabilities of the ESP-32 can be utilized for wireless monitoring and control. This would allow real-time data transmission, remote monitoring, and integration with mobile applications.

Integration of Machine Learning Algorithms

Machine learning techniques can be applied to EMG signal processing to improve accuracy and adaptability. Algorithms such as pattern recognition and neural networks can enable the system to learn user-specific muscle patterns and provide more precise control.

Application in Advanced Prosthetic Limbs

With further development, the system can be integrated into advanced prosthetic limbs with enhanced functionality, sensory feedback, and ergonomic design.

Miniaturization of Hardware

Future advancements in electronics can allow miniaturization of hardware components, making the system lighter, more comfortable, and suitable for wearable applications.

Chapter 5: Result and Conclusion

5.1 Result

The EMG controlled robotic arm was successfully designed, implemented, and tested under various operating conditions. The system demonstrated the ability to detect human muscle activity and translate it into mechanical movement of a robotic hand in real time.

During testing, the EMG sensor accurately detected muscle contractions when electrodes were properly placed on the user's arm. The output voltage of the EMG sensor increased proportionally with muscle contraction strength. This analog signal was successfully read by the ESP-32 microcontroller through its ADC without noticeable delay.

The ESP-32 processed the EMG signal effectively using threshold-based logic and generated appropriate PWM signals to control the servo motors. When the EMG signal exceeded the predefined threshold value, the servo motors rotated smoothly, causing the robotic hand fingers to close and grip objects. When the muscle relaxed, the servo motors returned to their initial positions, allowing the robotic hand to release the object.

The robotic hand was able to grip lightweight objects such as plastic cups, small tools, and cylindrical objects with reasonable stability. The response time of the system was found to be sufficiently fast for real-time operation, making the control feel natural and intuitive to the user.

Overall, the system achieved all the predefined objectives of the project and demonstrated reliable performance during testing.

EMG Signal Range (ADC Value)	Muscle Condition	Servo Motor Angle(°)	Robotic Hand Action
0 – 800	Relaxed muscle	0°	Hand fully open
800 – 1200	Slight contraction	20° – 70°	Partial finger movement
1200 – 1800	Moderate contraction	70° – 150°	Semi grip
1800 – 2400	Strong contraction	150° – 230°	Firm grip
Above 2400	Very strong contraction	240°	Maximum grip

Table : Relationship Between EMG Signal Strength and Servo Motor Angle

5.2 Discussion

The discussion section analyzes the observed results and evaluates the performance of the system in detail.

5.2.1 Findings

Several important findings were observed during the testing and operation of the EMG controlled robotic arm:

- The EMG sensor successfully detected muscle activity and produced a stable analog output when electrodes were correctly positioned.
- Threshold-based control was effective in distinguishing between relaxed and contracted muscle states.
- The ESP-32 microcontroller processed EMG signals efficiently and generated stable PWM signals.
- Servo motors responded accurately to PWM signals and provided smooth mechanical motion.
- The robotic hand performed gripping and releasing actions consistently under controlled conditions.
- Separate power supplies for the EMG sensor and servo motors significantly reduced signal noise and interference.

These findings confirm that EMG-based control is a practical and effective approach for robotic arm operation.

5.2.2 Analysis

Although the system performed successfully, several performance aspects were analyzed in detail. The accuracy of the system depended heavily on proper electrode placement and threshold tuning. Incorrect electrode positioning resulted in noisy signals and unstable operation.

The use of a single EMG sensor limited the system to basic gripping action rather than individual finger control. Additionally, muscle fatigue over extended use caused variations in EMG signal amplitude, which affected consistency.

Despite these limitations, the system demonstrated strong potential for assistive and educational applications. The use of ESP-32 provided sufficient processing power and flexibility, making the system scalable for future enhancements.

5.3 Conclusion

This project successfully demonstrated the design and implementation of an EMG Controlled Robotic Arm using ESP-32. The system effectively converted human muscle signals into mechanical motion, establishing a direct human–machine interface.

The integration of biomedical sensing, embedded system control, and robotic actuation was achieved using low-cost and easily available components. The project provided valuable practical experience in EMG signal acquisition, microcontroller programming, servo motor control, and system integration.

The results confirm that EMG-based control systems can be efficiently used in assistive technologies, rehabilitation devices, and prosthetic applications.

5.3.1 Recommendations

Based on the observations and analysis, the following recommendations are suggested for future improvement:

- Use multiple EMG sensors to achieve individual finger control.
- Apply advanced signal processing techniques to reduce noise.
- Integrate machine learning algorithms for adaptive thresholding.
- Add force feedback sensors for improved gripping control.
- Develop wireless monitoring using Wi-Fi or Bluetooth.
- Improve mechanical design for higher degrees of freedom.

These improvements would significantly enhance system performance and usability.

5.3.2 Final Conclusion

In conclusion, the EMG Controlled Robotic Arm using ESP-32 is a successful and meaningful engineering project that demonstrates the practical application of human muscle signals in robotic control. The system achieved real-time, intuitive, and reliable operation, fulfilling all project objectives.

This project highlights the potential of EMG-based control systems in modern engineering applications and serves as a strong foundation for future research and development in prosthetics, rehabilitation engineering, and human–robot interaction.

Appendix

```
#include <ESP32Servo.h>

Servo servo1;

Servo servo2;

// EMG analog input pin
const int emgPin = 34;

// Servo pins
const int servo1Pin = 13;
const int servo2Pin = 12;

// Filter variable
float smoothEMG = 0;

void setup() {
    Serial.begin(115200);

    // Attach servos
    servo1.attach(servo1Pin);
    servo2.attach(servo2Pin);

    // Set initial position
    servo1.write(0);
    servo2.write(0);

    delay(1000);
}
```

```
void loop() {  
  
    // Read EMG value (0–4095)  
    int raw = analogRead(emgPin);  
  
    // Smoothing filter (adjust if needed)  
    smoothEMG = (smoothEMG * 0.85) + (raw * 0.35);  
  
    // Convert EMG to servo range  
    // Adjust 800 → lower muscle level  
    // Adjust 2500 → strong muscle level  
    int angle = map(smoothEMG, 800, 2500, 0, 240);  
  
    // Limit movement  
    angle = constrain(angle, 0, 240);  
  
    // Move both servos  
    servo1.write(angle);  
    servo2.write(angle);  
  
    // Debug info  
    Serial.print("Raw EMG: ");  
    Serial.print(raw);  
    Serial.print(" Filtered: ");  
    Serial.print((int)smoothEMG);
```

```

Serial.print(" Angle: ");
Serial.println(angle);

// ===== Continuous EMG Output for Serial Plotter =====

Serial.print(raw);
Serial.print(",");
Serial.print((int)smoothEMG);
Serial.print(",");
Serial.println(angle);

delay(100);
}

```

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